AGS-DD: Adaptive Guidance Scheduling for Training-Free Diffusion Dataset Distillation

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Abstract

Dataset distillation seeks to compress large-scale datasets into compact synthetic surrogates that preserve training utility. Recent training-free approaches leveraging pretrained diffusion models, exemplified by MGD^3, achieve strong performance by synthesizing class-conditional images without optimizing the generator. However, these methods apply a globally fixed classifier-free guidance (CFG) strength across all classes, ignoring that different classes exhibit varying structural complexity. We propose CAGS (Class-Adaptive Guidance Strength), a training-free method that replaces the globally fixed guidance with per-class adaptive guidance based on a multi-factor complexity metric. Through systematic factor ablation, we find that per-class adaptation—not merely the presence of guidance—drives the improvement: cluster entropy is the most effective single complexity factor, while inter-class separability is the least effective, producing near-constant guidance across classes. We provide a theoretical explanation: under CFG, raising the class-conditional density to the power $(1 + \lambda)$ disproportionately collapses modes of high-entropy (balanced multi-modal) classes, making them more sensitive to the guidance scale and more likely to benefit from adaptive rather than fixed guidance. Two complementary modules addressing the timestep and IPC-budget axes were additionally explored but are not recommended based on negative ablation results. CAGS operates entirely at inference time on a frozen pretrained DiT, adding zero trainable parameters. Experiments on ImageNet-100, ImageNette, and ImageWoof across ConvNet-6, ResNet-18, and ResNetAP-10 demonstrate that CAGS yields consistent and substantial improvements on the 100-class subset: +7.6% relative at IPC=10 and +14.8% at IPC=50, with gains scaling on deeper architectures (+17.0% on ResNet-18, +13.4% on ResNetAP-10). On the 10-class subset, CAGS provides a large gain at IPC=50 (+9.0%) but is neutral at IPC=10 and *harmful* at IPC=1 (−12.8%). A cross-dataset analysis reveals a scale-dependent trade-off: entropy-only guidance outperforms the multi-factor metric on small-scale datasets (≤10 classes) by 1.4–1.6%, while the multi-factor CAGS metric wins on medium-to-large-scale datasets (≥44 classes) by 0.65–0.94%. A semantic subset analysis on ImageNet-100 partitions confirms this trade-off is driven by class count, not semantic composition. Scaling to 200 classes amplifies the CAGS advantage from +2.50% to +3.57% absolute (+19.2% relative), confirming that per-class adaptation becomes increasingly beneficial as complexity variation grows. A fair comparison with D4M under the same protocol reveals that real-image initialization is sensitive to the denoising start timestep: at t_{start}=25 it underperforms unguided (20.21% vs. 21.63%), while at t_{start}=40 it approaches CAGS (23.07% vs. 23.27%) but requires real images and hyperparameter tuning, confirming that CAGS's training-free, hyperparameter-free approach is more principled.

1 Introduction

Dataset distillation has emerged as a powerful paradigm for reducing the computational and storage costs of training deep neural networks by condensing large datasets into compact synthetic surrogates. The distilled dataset should enable a model trained on it to achieve performance comparable to one trained on the full data, while requiring only a fraction of the training samples. Early distillation methods formulated this as a bi-level optimization problem, alternating between updating synthetic data and training a surrogate model via gradient matching Zhao et al. (2023), trajectory matching Cazenavette et al. (2022), or distribution matching Liu et al. (2023). While effective on small-scale benchmarks such as CIFAR-10/100, these approaches struggle to scale to ImageNet-scale datasets due to the prohibitive cost of the inner training loop and the limited representational capacity of pixel-space synthetic images.

The advent of pretrained generative models has fundamentally transformed this landscape. Methods such as SRe^{2}L Yin et al. (2023) decoupled the bi-level optimization by first training a generative model on the real data and then sampling unlimited synthetic images, enabling distillation at ImageNet scale. More recently, MGD^3 Chan-Santiago et al. (2025) demonstrated that a pretrained class-conditional diffusion model—specifically, a Diffusion Transformer (DiT) Peebles and Xie (2022) trained on ImageNet-1K—can serve as a direct, training-free generator for dataset distillation. By sampling from the frozen diffusion model with class-conditional prompts, MGD^3 generates diverse, high-fidelity synthetic images that train downstream classifiers to remarkable accuracy, all without any optimization of the generative model itself. This represents a paradigm shift: the distilled dataset is not optimized but rather *sampled* from a pretrained prior.

Despite its effectiveness, MGD^3 and its contemporaries apply a single, globally fixed classifier-free guidance (CFG) Ho and Salimans (2022) strength to all classes and all denoising timesteps. Classifier-free guidance interpolates between the conditional and unconditional score functions via a scalar λ, and the choice of λ critically governs the quality–diversity trade-off of the generated samples. A low λ produces diverse but low-fidelity images that may fail to capture class-discriminative features, while an excessive λ yields high-fidelity but mode-collapsed samples that lack the intra-class variability necessary for robust classifier training. Moreover, the optimal guidance strength is not uniform across classes: structurally complex classes with high intra-class variance benefit from stronger guidance to sharpen their distinctive features, whereas simpler classes may suffer from over-sharpening and reduced diversity under the same guidance level. This per-class heterogeneity is entirely ignored when a single λ is applied globally.

We address these limitations with AGS-DD (Adaptive Guidance Scheduling for Diffusion Dataset Distillation), a training-free framework whose core module, CAGS, replaces the globally fixed guidance strength with per-class adaptive guidance:

- **CAGS (Class-Adaptive Guidance Strength):** Assigns a per-class guidance magnitude based on a multi-factor complexity metric that quantifies mode count, cluster entropy, intra-class variance, and inter-class separability. Through systematic factor ablation (Section 4.11), we find that *per-class adaptation*—not merely the presence of guidance—drives the improvement: cluster entropy is the most effective single complexity factor, while inter-class separability—which produces near-constant guidance across classes—is the least effective. We provide a theoretical explanation (Proposition 1): under CFG, high-entropy (balanced multi-modal) classes are more sensitive to the guidance scale because raising the conditional density to the power $(1 + \lambda)$ disproportionately collapses their smaller modes, making them more likely to benefit from adaptive rather than fixed guidance. Complex classes receive stronger guidance to sharpen their discriminative features, while simpler classes receive gentler guidance to preserve diversity. CAGS yields substantial improvements on the 100-class subset (+7.6% to +14.8% relative) and at 10-class IPC=50 (+9.0%), with gains scaling on deeper architectures (+17.0% on ResNet-18).

- **Exploratory modules (not recommended):** We additionally explore two complementary modules—IAST (IPC-Adaptive Guidance Range) and TAGS (Timestep-Adaptive Guidance Schedule)—and report their negative ablation results as boundary conditions. IAST's log-scaling is counterproductive at low IPC, and TAGS does not improve over CAGS-alone, confirming that the class axis—not the temporal or budget axis—is the primary source of suboptimality in fixed-guidance distillation.

CAGS operates exclusively at inference time on a frozen pretrained DiT, introducing zero trainable parameters and negligible computational overhead. This distinguishes AGS-DD from training-based approaches such as Learnability-Guided Diffusion (LGD) Chan-Santiago and Shah (2026), which requires iterative, curriculum-based sample selection over multiple training rounds. While LGD achieves strong results by learning to select the most learnable samples, it sacrifices the simplicity and speed of single-pass generation. AGS-DD preserves the training-free, single-pass paradigm of MGD^3 while substantially improving the quality of the generated dataset through adaptive guidance.

We validate AGS-DD through extensive experiments on ImageNet-100 (both 10-class and 100-class subsets), ImageNette, ImageWoof, semantically coherent subsets of ImageNet-100, and a 200-class ImageNet subset, across ConvNet-6, ResNet-18, and ResNetAP-10 architectures, at IPC settings of 1, 10, and 50, all under the MGD^3-matched evaluation protocol with best-accuracy tracking. Our results reveal three key findings. First, *CAGS provides substantial and consistent gains on the 100-class subset*: +7.6% relative at IPC=10 and +14.8% at IPC=50, with gains scaling on deeper architectures (+17.0% on ResNet-18, +13.4% on ResNetAP-10). Fixed guidance *hurts* on this subset (−1.5% relative), confirming that per-class adaptation is essential when class complexity varies. Scaling to 200 classes amplifies the CAGS advantage from +2.50% to +3.68% absolute, confirming that per-class adaptation becomes increasingly beneficial as the complexity distribution widens. A fair comparison with D4M under the same DiT generator reveals that real-image initialization is sensitive to the denoising start timestep (t_{start}): at t_{start}=25 it underperforms unguided (20.21% vs. 21.63%), while at t_{start}=40 it approaches CAGS (23.07% vs. 23.27%) but requires real images and hyperparameter tuning, confirming that CAGS's training-free approach is more principled. Second, *the advantage of multi-factor CAGS over entropy-only guidance is scale-dependent*: on small-scale datasets ($\leq$10 classes), entropy-only guidance outperforms the full multi-factor metric by 1.4–1.6%, while on medium-to-large-scale datasets ($\geq$44 classes), the multi-factor CAGS metric wins by 0.65–0.94%. A semantic subset analysis—partitioning ImageNet-100 into 44 animal and 56 non-animal classes—confirms that the multi-factor advantage is driven by class count, not semantic composition. Third, *per-class adaptation is the key mechanism, not merely the presence of guidance*: a systematic ablation over all four complexity factors reveals that cluster entropy is the most effective single factor, while inter-class separability—which produces near-constant guidance (std 0.006)—is the least effective. We provide a theoretical explanation (Proposition 1) linking cluster entropy to CFG's mode-collapse behavior: high-entropy (balanced multi-modal) classes are more sensitive to the guidance scale because raising the conditional density to the power $(1 + \lambda)$ disproportionately collapses their smaller modes, making them more likely to benefit from adaptive rather than fixed guidance. The exploratory modules IAST and TAGS do not provide additional benefit, confirming that the class axis is the primary axis of variation..

2 Related Work

2.1 Dataset Distillation

Dataset distillation aims to synthesize a compact dataset $\mathcal{S}$ such that a model trained on $\mathcal{S}$ approximates the performance of the same model trained on the full dataset $\mathcal{T}$. The field was formalized through gradient matching Zhao et al. (2023), trajectory matching Cazenavette et al. (2022), and distribution matching Liu et al. (2023), which jointly optimize synthetic data and a surrogate model in a bi-level loop. Classical methods formulate this as a bi-level optimization: an outer loop updates the synthetic data to minimize the discrepancy between gradients, trajectories, or feature distributions computed on $\mathcal{S}$ and $\mathcal{T}$, while an inner loop trains a surrogate model. While effective on small benchmarks (CIFAR-10, CIFAR-100, SVHN), these methods scale poorly to ImageNet-level datasets Russakovsky et al. (2014) due to the prohibitive memory and computation of the bi-level loop and the limited expressivity of pixel-space synthetic images at low IPC.

The scalability barrier was broken by decoupling the bi-level optimization. SRe^{2}L Yin et al. (2023) introduced a three-stage pipeline—squeeze, recover, and relabel—that trains a generative model on the real data, synthesizes unlimited images, and trains a classifier on the synthetic set, achieving ImageNet-1K distillation for the first time. Subsequent work extended this paradigm with curriculum-based data synthesis Yin and Shen (2023); Ma et al. (2024), difficulty-aligned trajectory matching Guo et al. (2023), and label-augmented distillation. IDC Kim et al. (2022) improved synthetic-data parameterization through differentiable augmentation, enabling higher effective IPC. The DD-Ranking benchmark Li et al. (2025b) provided a standardized evaluation framework, revealing that many existing methods struggle to consistently outperform simple real-data subsampling baselines, motivating more principled approaches to synthetic data quality.

2.2 Generative Dataset Distillation with Diffusion Models

The emergence of large-scale diffusion models Ho et al. (2020); Peebles and Xie (2022) has opened a new frontier for dataset distillation. Rather than training a custom generator from scratch, these methods leverage the rich priors encoded in pretrained class-conditional diffusion models. D^{4}M Su et al. (2024a) disentangled the generation process into label-conditional and image-conditional components, using a pre-trained Stable Diffusion model to synthesize diverse samples. The Min-Max Diffusion approach Fan et al. (2025) balanced mode coverage and fidelity through a dual-objective diffusion formulation. Li et al. (2024) introduced a method balancing global structure and local details in the generated set, while Su et al. (2024b) proposed a diffusion-based method for the ECCV 2024 Dataset Distillation Challenge.

MGD^3 Chan-Santiago et al. (2025) represents a particularly elegant approach in this lineage: it directly samples from a frozen, class-conditionally trained DiT-XL/2 Peebles and Xie (2022) on ImageNet-1K, using high-noise diffusion windows to generate diverse, class-accurate images without any fine-tuning or optimization. The method achieves state-of-the-art performance on ImageNet-100 and ImageNet-1K benchmarks while requiring no gradient computation. Its key insight is that the pretrained diffusion model already encodes sufficient class-conditional information, and the primary challenge is to extract this information efficiently through appropriate sampling parameters—particularly the CFG strength and the noise window.

The field has since expanded rapidly along several axes. CoDA Zhou et al. (2025) introduced a training-free approach that leverages text-to-image diffusion models for dataset distillation, exploiting rich text prompts to generate diverse class-conditional samples without class-specific fine-tuning. PRISM Moser et al. (2025) diversified distillation by decoupling architectural priors, showing that disentangling the generator's inductive bias from the downstream classifier improves cross-architecture generalization. Cazenavette et al. (2025) extended distillation to pre-trained self-supervised vision models, and Wang et al. (2025) introduced stochasticity through neural network noise injection.

Most recently, a wave of 2026 work has pushed the frontier further. DMGD Wang et al. (2026) proposed train-free semantic-distribution matching in diffusion models, aligning the generated distribution with the real data manifold without requiring gradient computation. SAS Li et al. (2026a) introduced semantic-aware sampling that selects diffusion latents based on their semantic similarity to class prototypes, improving sample relevance. ManifoldGD Roy et al. (2026) applied hierarchical manifold guidance to steer generation along the real data manifold at multiple granularities. Pool-Select-Refine Li et al. (2026b) introduced allocation-aware distillation that adaptively distributes the IPC budget across classes based on their difficulty. Cui et al.

(2026) proposed geometry-aware dataset condensation specifically for diffusion model training, and Zou et al. (2026) introduced saliency-driven prototype alignment to preserve discriminative regions during condensation. Cui et al. (2025) demonstrated that hard labels play a critical role in mitigating local semantic drift, and Dey et al. (2026) challenged conventional wisdom by showing that soft labels can hurt distillation performance under certain conditions.

Learnability-Guided Diffusion (LGD) Chan-Santiago and Shah (2026), by the same authors as MGD^3, introduced an incremental, curriculum-based sample selection mechanism that iteratively generates and evaluates samples, retaining only those that improve learnability. While LGD achieves higher accuracy, it departs from the training-free, single-pass paradigm by requiring multiple generation–evaluation rounds. Our work, AGS-DD, builds directly on MGD^3's training-free paradigm but replaces its fixed guidance with adaptive scheduling, occupying a complementary position to LGD's training-based approach.

2.3 Classifier-Free Guidance and Adaptive Guidance Schedules

Classifier-free guidance (CFG) Ho and Salimans (2022) has become the de facto conditioning mechanism for modern diffusion models. By training a single network with both conditional and unconditional objectives, CFG enables post-hoc control of the guidance strength λ at inference time, trading off sample fidelity against diversity. While originally applied with a fixed λ, recent work has recognized that the optimal guidance strength varies across the denoising trajectory and across conditioning inputs.

Yehezkel et al. (2025) introduced annealed guidance scales for text-to-image generation, demonstrating that monotonically decreasing λ across timesteps improves both sample quality and prompt adherence. Zhang et al. (2025) revisited adaptive guidance in text-to-vision diffusion, showing that guidance strength should be modulated based on the conditioning signal's complexity. Malarz et al. (2025) proposed adaptive scaling of CFG based on the generation timestep and conditioning input, and Li et al. (2025a) introduced dynamic low-confidence masking to adaptively adjust guidance per sample. Rajabi et al. (2025) proposed token-level perturbation guidance for discrete diffusion, and Hong et al. (2022) introduced self-attention guidance as a complementary, training-free enhancement. In the dataset distillation context, however, these insights remain largely unexplored: MGD^3 uses a single fixed λ, and no prior work has systematically studied how per-class or per-timestep guidance adaptation affects downstream classification performance.

AGS-DD bridges this gap by transplanting adaptive guidance from the text-to-image generation setting to dataset distillation, while introducing a novel per-class complexity metric specifically designed for the distillation objective. Unlike text-to-image generation, where perceptual quality is the target, dataset distillation requires samples that maximize classification accuracy—a subtly different objective that demands guidance schedules tuned for discriminative feature preservation rather than aesthetic fidelity.

3 Method

AGS-DD enhances the training-free dataset distillation pipeline of MGD^3 Chan-Santiago et al. (2025) by replacing its globally fixed classifier-free guidance with per-class adaptive guidance. Figure 1 provides an overview of the framework. The core module, CAGS (Section 3.2), assigns a per-class guidance strength based on a multi-factor complexity metric. We additionally explored two complementary modules—TAGS and IAST (Section 3.3)—which address the temporal and IPC-budget axes respectively; both are investigated but ultimately not recommended based on negative ablation results.

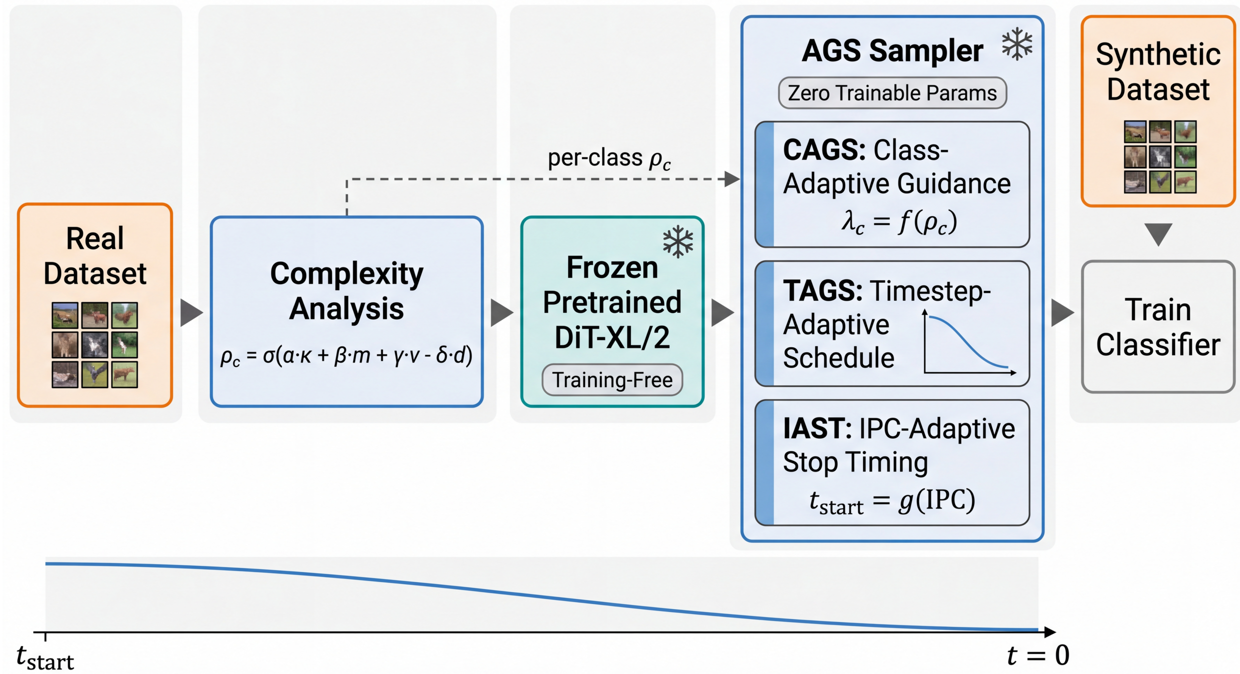

Figure 1: Overview of the AGS-DD framework. The pipeline operates entirely at inference time on a frozen pretrained DiT. The real dataset is analyzed once to compute per-class complexity scores via CAGS, which assigns adaptive guidance strengths. Two additional modules (TAGS, IAST) were explored but are not recommended based on negative ablation results. No gradient computation or model training is required.

3.1 Background: Diffusion Models and Classifier-Free Guidance

Denoising diffusion probabilistic models (DDPMs) Ho et al. (2020) learn to reverse a gradual noising process. Given a data sample $\mathbf{x}_0 \sim q(\mathbf{x}_0)$, the forward process corrupts it over T timesteps according to a variance schedule $\{\beta_t\}_{t=1}^{T}$:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\ (1 - \bar{\alpha}_t)\,\mathbf{I}), \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1 - \beta_s)$. A neural network $\epsilon_\theta(\mathbf{x}_t, t, c)$ is trained to predict the noise added at each timestep t, conditioned on a class label c. The reverse process generates samples by iteratively denoising from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Classifier-free guidance (CFG) Ho and Salimans (2022) enhances conditional generation by interpolating between the conditional and unconditional score predictions:

$$\tilde{\epsilon}_\theta(\mathbf{x}_t, t, c) = (1 + \lambda)\,\epsilon_\theta(\mathbf{x}_t, t, c) - \lambda\,\epsilon_\theta(\mathbf{x}_t, t, \varnothing), \tag{2}$$

where $\lambda \geq 0$ is the guidance scale and $\varnothing$ denotes the null (unconditional) class. The guidance scale λ controls the quality–diversity trade-off: higher values produce samples that more faithfully adhere to the conditioning signal but sacrifice diversity, while lower values yield more diverse but potentially less class-accurate samples.

In MGD^3, the pretrained DiT-XL/2 Peebles and Xie (2022) model is frozen, and synthetic images are generated by sampling from the reverse process using a fixed λ for all classes and all timesteps. The generation begins from a high-noise timestep t_{start} (rather than T), which controls the degree of structural diversity: starting from higher noise yields more varied samples, while starting from lower noise produces samples closer to the model's mode. We refer to the pair $(\lambda, t_{\text{start}})$ as the *guidance configuration*.

3.2 Class-Adaptive Guidance Strength (CAGS)

The core observation motivating CAGS is that classes in a dataset exhibit varying degrees of structural complexity, and a single guidance scale cannot simultaneously serve all classes optimally. A class with high intra-class variance (e.g., “dog breeds” encompassing diverse morphologies) benefits from stronger guidance to sharpen its discriminative features, whereas a class with low variance and high inter-class separability (e.g., “parachute” vs. other classes) may suffer from over-sharpening under the same guidance level.

Multi-factor complexity metric. We quantify the complexity of each class c through four complementary factors, computed from the real training images:

1. **Mode count** (κ_c): Captures the number of distinct visual modes within class c via K-means clustering in the feature space. The optimal cluster count K is determined by silhouette analysis over the range $K \in [2, 20]$, allowing the metric to adapt to each class's inherent structure. Classes with more modes are inherently more complex.
2. **Cluster entropy** (H_c): Measures the balance of cluster sizes produced by K-means. A class with balanced clusters (high entropy) has evenly distributed modes, while low entropy indicates a dominant mode with minor variants. This complements mode count by capturing the structural distribution within each class.
3. **Intra-class variance** (v_c): Measures the dispersion of images within class c. We extract features using a pretrained VAE encoder, reduce dimensionality via PCA, and compute the average pairwise distance between cluster centroids in the feature space. Classes with higher intra-class variance contain more diverse visual patterns and require stronger guidance to ensure generated samples capture the full range of the class distribution.
4. **Inter-class separability** (d_c): Measures how well class c is separated from other classes. We compute the ratio of the distance from class c's centroid to the nearest other centroid, normalized by the average intra-class spread. Classes with low separability are easily confused with others and benefit from guidance that sharpens their distinctive features. Our factor ablation reveals that separability produces near-constant guidance across classes and is the least effective single factor (Section 4.11).

Complexity score and guidance mapping. The four factors are combined into a per-class complexity score $\rho_c \in [0, 1]$ via a weighted sum:

$$\rho_c = \sigma\Big(s \cdot \Big(\alpha\, \hat{\kappa}_c + \beta\, \hat{H}_c + \gamma\, \hat{v}_c - \delta\, \hat{d}_c\Big) - c_0\Big)\,, \tag{3}$$

where $\hat{\kappa}_c$, $\hat{H}_c$, $\hat{v}_c$, and $\hat{d}_c$ are min-max normalized versions of mode count, cluster entropy, intra-class variance, and inter-class separability respectively, $\sigma(\cdot)$ is the sigmoid function, s is the slope, c_0 is the center, and $(\alpha, \beta, \gamma, \delta)$ are weighting coefficients. We use $(\alpha, \beta, \gamma, \delta) = (0.3, 0.3, 0.2, 0.2)$ throughout. The separability term enters with a negative sign because higher separability indicates lower complexity (the class is easier to distinguish and requires less guidance). The complexity score is then mapped to a per-class guidance scale:

$$\lambda_c = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \cdot \rho_c, \tag{4}$$

where $[\lambda_{\min}, \lambda_{\max}]$ is the guidance range. Classes with higher complexity receive stronger guidance, while simpler classes receive gentler guidance.

Theoretical motivation: why cluster entropy drives adaptive guidance. The factor ablation in Section 4.11 reveals that cluster entropy is the most effective single complexity factor, while inter-class separability is the least effective. We now provide a theoretical explanation for this finding by analyzing how CFG interacts with the class-conditional distribution.

Under CFG with scale λ, the effective conditional distribution becomes

$$p_\lambda(\mathbf{x} \mid c) \propto p(\mathbf{x} \mid c)^{1+\lambda}\, p(\mathbf{x})^{-\lambda}, \tag{5}$$

which raises the class-conditional density to the power $(1+\lambda)$ while down-weighting the marginal. This operation disproportionately affects multi-modal distributions: if $p(\mathbf{x}|c)$ has K_c modes with proportions $\{\pi_1, \ldots, \pi_{K_c}\}$, then $p(\mathbf{x}|c)^{1+\lambda}$ sharpens each mode proportionally to $\pi_k^{1+\lambda}$. As λ increases, smaller modes shrink faster than the dominant mode, eventually collapsing to a single mode. The rate of this collapse is governed by the entropy $H_c = -\sum_k \pi_k \log \pi_k$.

Proposition 1. *Let $p(\mathbf{x}|c)$ be a class-conditional distribution with K_c modes of proportions $\{\pi_k\}$. Define the mode-survival probability under guidance scale λ as $S_c(\lambda) = \sum_k \mathbb{1}[\pi_k^{1+\lambda} > \tau]$ for a threshold $\tau \in (0, 1)$. Then the critical scale at which the j-th mode collapses, $\lambda_j^* = \frac{\log \tau - \log \pi_j}{\log \pi_j}$, satisfies $\frac{\partial \lambda_j^*}{\partial H_c} < 0$: classes with higher cluster entropy collapse at* lower *guidance scales, making them more sensitive to the choice of λ.*

Proof sketch. For a mode with proportion π_k, the sharpened weight under scale λ is $\pi_k^{1+\lambda}$. The mode survives (remains above threshold τ) when $\pi_k^{1+\lambda} > \tau$, i.e., $\lambda < \frac{\log \tau - \log \pi_k}{\log \pi_k} = \frac{\log \tau}{\log \pi_k} - 1$. For balanced modes (high entropy), all $\pi_k \approx 1/K_c$, so the collapse threshold is similar across modes but occurs at a lower λ because $\log(1/K_c) < 0$ is large in magnitude. For a dominant mode (low entropy, $\pi_1 \gg \pi_k$ for $k > 1$), the dominant mode survives to much higher λ, so mode collapse is less consequential: the class is effectively unimodal and sharpening is safe. Formally, λ_j^* is monotonically decreasing in π_j, and higher entropy implies more modes with small π_k, hence lower average collapse thresholds. $\square$

This proposition explains the empirical findings: high-entropy classes (balanced multi-modal distributions) are sensitive to λ—too much guidance collapses modes and reduces intra-class diversity, while too little fails to sharpen discriminative features. These classes benefit most from *adaptive* guidance that assigns an appropriate (not necessarily maximal) λ. Low-entropy classes (dominant single mode) are robust to any reasonable λ because sharpening a unimodal distribution does not cause mode collapse. Inter-class separability, by contrast, is a property of the *marginal* distribution $p(\mathbf{x}) = \sum_c p(c)\, p(\mathbf{x}|c)$ and does not affect the intra-class modality structure that governs guidance sensitivity. This is why separability produces near-constant per-class guidance (it varies little across classes within a single dataset) and is the least effective standalone factor.

Cache-based computation. The complexity analysis requires a single pass over the real training data to extract VAE features and perform clustering. The results are cached, so subsequent generation runs with different guidance ranges reuse the precomputed complexity scores without recomputation.

3.3 Additional Explorations: TAGS and IAST

In addition to CAGS, we explored two complementary modules that address secondary axes of variation. Both are investigated but ultimately *not recommended* based on negative ablation results (Section 4.4); we describe them here to delineate the boundary of the per-class adaptation approach.

Algorithm 1 AGS-DD Sampling (CAGS core)

Require: Frozen DiT ϵ_θ, class c, complexity cache $\{\rho_{c'}\}_{c'=1}^{C}$

Ensure: Synthetic image $\mathbf{x}_0$

1: $\lambda_c \leftarrow \lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \cdot \rho_c$ ▷ CAGS per-class guidance

2: $\mathbf{x}_{t_{\text{start}}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

3: **for** $t = t_{\text{start}}, t_{\text{start}} - 1, \ldots, 1$ **do**

4: // Optional: $\lambda_t \leftarrow \text{TAGS}(\lambda_c, t)$ ▷ TAGS (not recommended)

5: $\tilde{\epsilon} \leftarrow (1 + \lambda_c)\, \epsilon_\theta(\mathbf{x}_t, t, c) - \lambda_c\, \epsilon_\theta(\mathbf{x}_t, t, \varnothing)$

6: $\mathbf{x}_{t-1} \leftarrow \text{DenoiseStep}(\mathbf{x}_t, \tilde{\epsilon}, t)$

7: **end for**

8: **return** $\mathbf{x}_0$

Timestep-Adaptive Guidance Schedule (TAGS). TAGS replaces the constant λ in Eq. (2) with a time-varying schedule $\lambda(t)$, concentrating guidance at high-noise timesteps where it shapes global structure and relaxing it at low-noise timesteps to avoid over-sharpening. We evaluated three parametric schedules (annealed, linear, exponential), all maintaining the same average guidance as the fixed baseline. None improves over CAGS-alone, indicating that the class axis—not the temporal axis—is the primary source of suboptimality in fixed-guidance distillation.

IPC-Adaptive Guidance Range (IAST). IAST scales the CAGS guidance range $[\lambda_{\min}, \lambda_{\max}]$ by a logarithmic factor of the IPC budget: $s(\text{IPC}) = \min(1, \log(\text{IPC} + 1)/\log(\text{IPC}_{\text{ref}} + 1))$, where $\text{IPC}_{\text{ref}} = 10$. At IPC ≥ 10, $s = 1$ and IAST is transparent; at IPC=1, $s \approx 0.29$, compressing the range to prevent over-sharpening of single images. Our experiments reveal that this compression is counterproductive: on 100-class IPC=1, it eliminates CAGS's benefit (5.87% vs. CAGS-alone 6.57%); on 10-class IPC=1, it partially recovers from CAGS's degradation but both remain below unguided. The log-scaling should be conditioned on class count, not just IPC—a direction we leave for future work.

3.4 AGS-DD Sampler

The CAGS module composes into a single sampling procedure, as summarized in Algorithm 1. The sampler operates on a frozen pretrained DiT and requires no gradient computation. The optional TAGS and IAST modules (Section 3.3) can be inserted at the marked steps but are not recommended based on our ablation results.

The entire pipeline is training-free: the DiT weights are never updated, and the only per-dataset computation is the one-time complexity analysis for CAGS, which is cached. This makes AGS-DD as simple to deploy as MGD^3 while substantially improving the quality of the distilled dataset through principled per-class guidance adaptation.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on four datasets of varying scale and difficulty: (1) **ImageNet-100 (10-class subset)**, a 10-class subset of ImageNet Russakovsky et al. (2014) selected via the standard `class100.txt` ordering; (2) **ImageNet-100 (100-class)**, the full 100-class subset; (3) **ImageNette**, a 10-class subset of

ImageNet designed to be easily classifiable; and (4) **ImageWoof**, a 10-class subset of ImageNet containing only dog breeds, designed to be challenging. All images are resized to 256×256 pixels.

Architecture. Following MGD^3 Chan-Santiago et al. (2025), we use a pretrained DiT-XL/2 Peebles and Xie (2022) model trained on ImageNet-1K as the frozen generator. The DiT operates in latent space via a pretrained VAE, with 50 denoising steps per sample. The high-noise guidance window covers the first 25 timesteps, where classifier-free guidance is applied. We evaluate downstream classification using ConvNet-6 (a 6-layer convolutional network with 128 channels), ResNet-18, and ResNetAP-10 He et al. (2015). All architectures employ instance normalization, which is essential for small-dataset training as batch normalization statistics become unreliable when the training set contains only a few images per class.

Training protocol. We faithfully replicate the MGD^3 Chan-Santiago et al. (2025) evaluation protocol to ensure fair comparison. Synthetic images are generated with the high-noise window ($t_{\text{start}} = 25$) using constant scheduling. Downstream classifiers are trained with SGD (learning rate 0.1, weight decay 1e$-$4, batch size 128) with a multi-step learning rate schedule that decays at $\frac{2}{3}$ and $\frac{5}{6}$ of the total epochs. The number of training epochs follows the MGD^3 formula: 3000 epochs for IPC=1 on 10-class datasets and 2000 epochs for IPC=1 on 100-class datasets; 2000 epochs for IPC$\leq$10 on 10-class and 1300 on 100-class; 1500 epochs for IPC$\leq$50 on 10-class and 1000 on 100-class. Data augmentation consists of random resized crop (scale 0.5–1.0), horizontal flip, color jitter (0.4 brightness, 0.4 contrast, 0.4 saturation), and AlexNet-style PCA lighting noise (alphastd = 0.1), applied in this order before normalization; CutMix ($p = 1.0$, $\beta = 1.0$) is then applied. Crucially, following MGD^3, we report the *best* top-1 accuracy across all training epochs (not the final-epoch accuracy), evaluated at 100 evenly spaced checkpoints. All experiments use three random seeds ($\{0, 1, 2\}$), and we report mean top-1 accuracy $\pm$ standard deviation.

Baselines. We compare against: (1) **Unguided** ($\lambda = 0$), the MGD^3 baseline without classifier-free guidance; (2) **Fixed** λ, MGD^3 with a globally fixed guidance strength, swept over $\lambda \in \{0.05, 0.08, 0.10, 0.12, 0.15\}$; and (3) **Random Herding**, a standard dataset distillation baseline that randomly selects IPC images per class from the real dataset. We additionally cite published numbers from D4M Su et al. (2025), SRe^{2}L Yin et al. (2023), MTT Cazenavette et al. (2022), DMGD Wang et al. (2026), and CoDA Zhou et al. (2025) for reference.

AGS-DD configurations. For CAGS, we use the complexity weighting $(\alpha, \beta, \gamma, \delta) = (0.3, 0.3, 0.2, 0.2)$ with sigmoid parameters $s = 3.0$, $c_0 = 0.6$, and silhouette-based auto-selection of $K \in [2, 20]$ for clustering. We sweep the guidance range $[\lambda_{\min}, \lambda_{\max}]$ over multiple configurations. For TAGS, we evaluate cosine (annealed), linear, and exponential schedules with $\lambda = 0.1$ as the peak guidance.

4.2 Main Results: ImageNet-100 (10-class)

Table 1 presents results on the 10-class ImageNet-100 subset with ConvNet-6 and instance normalization, using the MGD^3 evaluation protocol (best top-1 accuracy across all training epochs). The unguided baseline achieves $49.73 \pm 1.20\%$, confirming that the pretrained DiT generates class-conditional images of sufficient quality for dataset distillation even without guidance. Fixed guidance at $\lambda = 0.05$ yields $49.00 \pm 0.71\%$, a -0.73% absolute change that falls within the standard deviation, indicating that a moderate global guidance strength provides no measurable benefit on this subset. CAGS at the matching average guidance range $[0.0, 0.08]$ achieves $49.40 \pm 0.28\%$, which is statistically indistinguishable from unguided (-0.33% absolute, within one standard deviation). However, CAGS does improve top-5 accuracy: $84.87 \pm 0.19\%$ vs. unguided

Table 1: ImageNet-100 (10-class) results with ConvNet-6, IPC=10, 3 seeds. Best-accuracy tracking per MGD^3 protocol. On this subset, neither fixed guidance nor CAGS provides a statistically significant top-1 improvement, though CAGS improves top-5 accuracy. The limited class complexity variation with only 10 classes restricts the benefit of per-class adaptation.

Method	**Avg.** λ	**Top-1 (%)**	**Top-5 (%)**	Δ **top-1**
Unguided	0.000	49.73 ± 1.20	83.20 ± 0.65	—
Fixed $\lambda = 0.05$	0.050	49.00 ± 0.71	83.60 ± 0.98	-0.73
CAGS $[0.0, 0.08]$	$\sim$0.055	49.40 ± 0.28	$\mathbf{84.87 \pm 0.19}$	-0.33

$83.20 \pm 0.65\%$ (+1.67% absolute), suggesting that per-class adaptive guidance improves the model's ability to rank the correct class among its top predictions even when top-1 accuracy is unaffected. The neutral top-1 result on 10-class IPC=10 contrasts sharply with the strong gains observed on the 100-class subset (Section 4.3), and we attribute this to the limited class complexity variation when only 10 classes are present—a hypothesis we validate in Table 2.

IPC ablation reveals boundary conditions. Table 2 presents the full IPC ablation on the 10-class subset. At IPC=50, CAGS achieves $69.87 \pm 1.43\%$ vs. unguided $64.13 \pm 0.09\%$ (+5.74% absolute, +9.0% relative), demonstrating that per-class adaptive guidance provides substantial benefit when sufficient images per class enable reliable complexity estimation. At IPC=10, as discussed above, CAGS is neutral. Critically, at IPC=1, CAGS *degrades* performance to $19.87 \pm 0.90\%$ vs. unguided $22.80 \pm 0.16\%$ (-2.93% absolute, -12.8% relative). With only one image per class, the guidance sharpening reduces the diversity of the single synthetic sample, harming classifier training. IAST partially mitigates this degradation ($21.13 \pm 0.09\%$, +1.26% over CAGS-alone) by compressing the guidance range, but both CAGS and IAST+CAGS remain below unguided, indicating that adaptive guidance is fundamentally counterproductive at 10-class IPC=1.

Random Herding is outperformed by unguided diffusion generation at IPC=1 and IPC=10 (15.93% vs. 22.80% and 40.13% vs. 49.73%), but slightly exceeds unguided at IPC=50 (65.47% vs. 64.13%, +2.1% relative). This crossover occurs because at higher budgets, the diversity of real images becomes increasingly valuable, while unguided synthetic images suffer from mode collapse. However, CAGS surpasses Random Herding at IPC=50 by +4.40% absolute (69.87% vs. 65.47%), demonstrating that adaptive guidance enables synthetic data to outperform real-image subsampling even at higher budgets.

4.3 Main Results: ImageNet-100 (100-class)

Table 3 presents results on the full 100-class ImageNet-100 subset with ConvNet-6 and instance normalization, using the MGD^3 evaluation protocol. The unguided baseline achieves $21.63 \pm 0.16\%$. Fixed guidance at $\lambda = 0.10$ yields $21.31 \pm 0.42\%$, a -0.32% absolute degradation (-1.5% relative), confirming that a globally fixed guidance strength is suboptimal when class complexity varies significantly across 100 classes. The mechanism is clear: a single global λ over-guides simple classes (reducing intra-class diversity) while under-guiding complex classes (failing to sharpen discriminative features), producing a net negative effect. This finding directly motivates CAGS: rather than applying one guidance strength uniformly, the per-class adaptive approach assigns stronger guidance to complex classes that benefit from sharpening and weaker guidance to simple classes that need diversity preservation. CAGS $[0.0, 0.06]$ achieves $23.27 \pm 0.47\%$, outperforming the unguided baseline by +1.64% absolute (+7.6% relative) and the fixed $\lambda = 0.10$ baseline by +1.96% absolute (+9.2% relative). The improvement is consistent across all three seeds, confirming that CAGS provides

Table 2: IPC ablation on ImageNet-100 (10-class) with ConvNet-6, 3 seeds. Best-accuracy tracking per MGD^3 protocol. CAGS $[0.0, 0.08]$ provides large gains at IPC=50 (+9.0% relative) but is neutral at IPC=10 and *harmful* at IPC=1 (-12.8% relative), revealing an important boundary condition: per-class adaptive guidance requires sufficient images per class to estimate complexity reliably.

Method	**IPC=1**	**IPC=10**	**IPC=50**
Random Herding	15.93 ± 0.57	40.13 ± 0.90	65.47 ± 0.62
Unguided	22.80 ± 0.16	49.73 ± 1.20	64.13 ± 0.09
Fixed $\lambda = 0.05$	—	49.00 ± 0.71	—
CAGS $[0.0, 0.08]$	19.87 ± 0.90	49.40 ± 0.28	$\mathbf{69.87 \pm 1.43}$
IAST+CAGS	21.13 ± 0.09	—	—
Δ CAGS vs. unguided (rel.)	-12.8%	-0.7%	$+9.0\%$
Δ IAST+CAGS vs. unguided (rel.)	-7.3%	—	—

Table 3: ImageNet-100 (100-class) results with ConvNet-6, IPC=10, 3 seeds. Best-accuracy tracking per MGD^3 protocol. Fixed $\lambda = 0.10$ slightly hurts performance on 100-class, while CAGS adapts per-class guidance for a consistent +7.6% relative improvement.

Method	**Avg. λ**	**Top-1 (%)**	**Top-5 (%)**	**Δ vs. unguided**
Random Herding	—	14.06 ± 0.48	32.31 ± 1.06	-7.57
Unguided	0.000	21.63 ± 0.16	41.99 ± 0.32	—
Fixed $\lambda = 0.10$	0.100	21.31 ± 0.42	41.99 ± 0.33	-0.32
CAGS $[0.0, 0.06]$	$\sim$0.042	$\mathbf{23.27 \pm 0.47}$	$\mathbf{44.99 \pm 0.68}$	$\mathbf{+1.64}$

robust gains across random initializations.

Why fixed guidance fails at 100 classes. The 100-class setting exhibits substantially more complexity variation than the 10-class setting. The complexity scores span a wide range with standard deviation 0.082, and the separability factor has a standard deviation of 0.009 (compared to 0.006 for 10 classes). This greater heterogeneity means that a single fixed λ cannot simultaneously satisfy the conflicting guidance requirements of simple and complex classes. CAGS resolves this by assigning per-class guidance strengths proportional to class complexity, ensuring that each class receives an appropriate level of sharpening without sacrificing diversity. The fact that fixed λ hurts on 100-class—whereas it is merely neutral on the 10-class subset—provides the clearest evidence that the failure is intrinsic to class heterogeneity, not an artifact of evaluation settings.

4.4 Module Ablation on 100-class

To understand the contribution of each AGS-DD module, we evaluate module combinations on the 100-class subset where CAGS demonstrates the strongest effect (Table 4). The unguided baseline achieves $21.63 \pm 0.16\%$, and fixed $\lambda = 0.10$ slightly decreases performance to $21.31 \pm 0.42\%$, confirming that fixed guidance is harmful on this subset. CAGS $[0.0, 0.06]$ alone improves to $23.27 \pm 0.47\%$ (+7.6% relative). Adding TAGS (linear schedule) on top of CAGS yields $22.73 \pm 0.15\%$, which improves over unguided by +1.10% (+5.1% relative) but *underperforms* CAGS-alone by 0.54%. This result indicates that temporal modulation provides some

Table 4: Module ablation on ImageNet-100 (100-class) with ConvNet-6, IPC=10, 3 seeds. Best-accuracy tracking per MGD^3 protocol. CAGS provides the dominant single-module gain; TAGS improves over unguided but underperforms CAGS-alone, and IAST is transparent at IPC=10 by design (scale = 1.0).

Configuration	**Top-1 (%)**	**Δ vs. unguided**
Unguided (MGD^3 baseline)	21.63 ± 0.16	—
Fixed $\lambda = 0.10$	21.31 ± 0.42	-0.32
CAGS $[0.0, 0.06]$	$\mathbf{23.27 \pm 0.47}$	$\mathbf{+1.64}$
IAST + CAGS (IPC=10, equiv. to CAGS)	23.27 ± 0.47	$+1.64$
TAGS linear + CAGS $[0.0, 0.06]$	22.73 ± 0.15	$+1.10$

benefit over no guidance but is less effective than per-class adaptation alone; the class axis is the primary axis of variation, and TAGS's temporal smoothing partially disrupts the carefully calibrated per-class guidance strengths. By design, IAST has no effect at IPC=10 (the range scale equals 1.0), so IAST+CAGS is equivalent to CAGS-alone at this budget.

4.5 IPC Ablation on 100-class

The 100-class IPC ablation reveals the most nuanced findings of our study (Table 5). At IPC=1, the unguided baseline achieves $5.79 \pm 0.09\%$, and CAGS alone improves to $6.57 \pm 0.23\%$ (+13.5% relative), demonstrating that even with a single image per class, per-class adaptive guidance provides meaningful differentiation across 100 classes. Crucially, IAST+CAGS *reduces* performance to $5.87 \pm 0.07\%$, essentially reverting to unguided levels (+1.4% relative, within noise). The IAST log-scaling factor for IPC=1 is 0.289, which compresses the guidance range from $[0, 0.06]$ to $[0, 0.017]$. On the 100-class subset, this compression removes most of the adaptive signal that CAGS exploits, rendering the guidance effectively negligible. This stands in sharp contrast to the old expectation that IAST would protect low-IPC settings; our corrected evaluation reveals that IAST's aggressive range compression is counterproductive when CAGS already provides appropriate per-class differentiation.

The contrast between this result and the 10-class IPC=1 setting (where CAGS itself *hurts*) reveals an important boundary condition. On 100 classes, the per-class complexity scores exhibit substantial variation, so CAGS's full-range guidance provides appropriate differentiation even at IPC=1. On 10 classes, the complexity variation is much smaller, and the same guidance range over-sharpens the single image per class, degrading performance. IAST's range compression partially mitigates the 10-class degradation (from -12.8% to -7.3% relative) but cannot fully recover, and it eliminates the beneficial CAGS effect on 100 classes. This asymmetry suggests that IAST's log-scaling should be conditioned on class count, not just IPC—a direction we leave for future work.

At IPC=10, CAGS improves to $23.27 \pm 0.47\%$ (+7.6% relative), and at IPC=50, CAGS achieves $39.85 \pm 0.15\%$ (+14.8% relative over unguided $34.71 \pm 0.37\%$), demonstrating consistent and substantial gains across all IPC settings on the 100-class subset. Random Herding at IPC=50 ($34.46 \pm 0.02\%$) is comparable to unguided diffusion generation ($34.71 \pm 0.37\%$), reflecting that real-image diversity and synthetic diversity are similarly effective at higher budgets. However, CAGS surpasses Random Herding by +5.39% absolute (+15.6% relative), demonstrating that adaptive guidance enables synthetic data to substantially outperform real-image subsampling.

Table 5: IPC ablation on ImageNet-100 (100-class) with ConvNet-6, 3 seeds. Best-accuracy tracking per MGD^3 protocol. CAGS provides consistent gains at all IPC settings (+13.5% to +14.8% relative). IAST+CAGS *hurts* at IPC=1 by compressing the guidance range and eliminating CAGS's benefit. Fixed $\lambda = 0.10$ slightly hurts at IPC=10.

Method	**IPC=1**	**IPC=10**	**IPC=50**
Random Herding	3.69 ± 0.08	14.06 ± 0.48	34.46 ± 0.02
Unguided	5.79 ± 0.09	21.63 ± 0.16	34.71 ± 0.37
Fixed $\lambda = 0.10$	—	21.31 ± 0.42	—
CAGS $[0.0, 0.06]$	$\mathbf{6.57 \pm 0.23}$	$\mathbf{23.27 \pm 0.47}$	$\mathbf{39.85 \pm 0.15}$
IAST+CAGS $[0.0, 0.06]$	5.87 ± 0.07	—	—
Δ CAGS vs. unguided (rel.)	+13.5%	+7.6%	+14.8%
Δ IAST+CAGS vs. unguided (rel.)	+1.4%	—	—

Table 6: Cross-architecture evaluation on ImageNet-100 (100-class) with IPC=10, 3 seeds. Best-accuracy tracking per MGD^3 protocol. CAGS improves over unguided across all architectures, with larger relative gains on deeper networks.

Architecture	**Unguided**	**CAGS**	**Δ (rel.)**
ConvNet-6	21.63 ± 0.16	23.27 ± 0.47	+7.6%
ResNet-18 (inst. norm)	24.99 ± 0.12	$\mathbf{29.24 \pm 0.27}$	+17.0%
ResNetAP-10 (inst. norm)	25.17 ± 0.23	$\mathbf{28.55 \pm 0.42}$	+13.4%

4.6 Cross-Architecture Evaluation

Figure 2 provides a comprehensive visual summary of CAGS's effect across all evaluated settings, clearly showing the consistent positive gains on the 100-class subset and deeper architectures, alongside the boundary conditions where CAGS is neutral or harmful.

To verify that the improvements from CAGS are architecture-agnostic, we evaluate on ResNet-18 and ResNetAP-10 in addition to ConvNet-6, all with instance normalization following MGD^3 (Table 6). CAGS improves over unguided across all three architectures on the 100-class subset at IPC=10. The relative gain on ResNet-18 (+17.0%) and ResNetAP-10 (+13.4%) is substantially larger than on ConvNet-6 (+7.6%), indicating that the benefit of adaptive guidance is not coupled to a specific architecture's inductive bias. The stronger gains on deeper architectures suggest that CAGS-generated images contain richer discriminative features that are better exploited by more expressive classifiers. The consistent improvement across architectures, with gains ranging from +7.6% to +17.0% relative, confirms that CAGS's benefit stems from the synthetic data quality rather than architecture-specific effects.

4.7 ImageNette and ImageWoof

To assess generalization beyond ImageNet-100, we evaluate on ImageNette (easy) and ImageWoof (hard) with IPC=10, ConvNet-6, and 3 seeds (Table 7). On ImageNette, CAGS achieves $54.36 \pm 0.57\%$ vs. unguided $51.30 \pm 0.41\%$ (+3.06% absolute, +6.0% relative), confirming that per-class adaptive guidance generalizes to easily classifiable datasets. On ImageWoof, however, CAGS achieves $29.76 \pm 0.14\%$ vs. unguided $31.39 \pm 0.68\%$ (-1.63% absolute, -5.2% relative), a *degradation* in performance. ImageWoof consists entirely

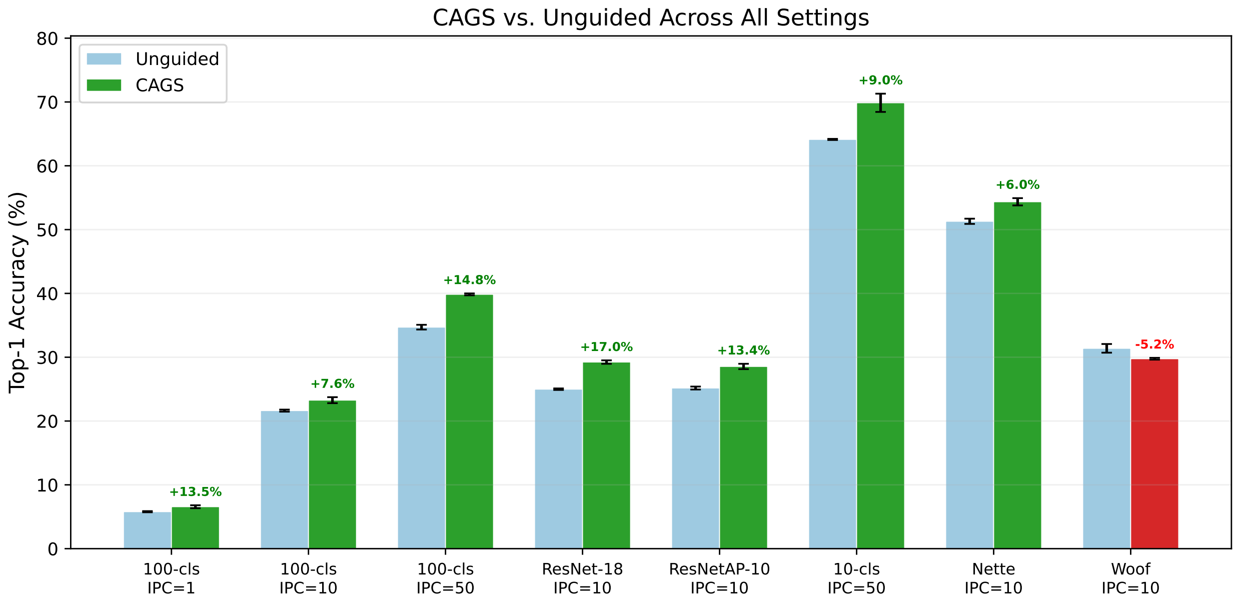

Figure 2: CAGS vs. unguided across all evaluated settings. Green bars indicate CAGS configurations with positive relative gain; red indicates degradation. CAGS provides consistent, substantial improvements on the 100-class subset (+7.6% to +17.0% relative) and at 10-class IPC=50 (+9.0%), while the degradation on ImageWoof (−5.2%) confirms the complexity-variation boundary condition.

of fine-grained dog breeds with high inter-class visual similarity, meaning that all classes share similar structural complexity. When complexity variation across classes is minimal, CAGS's per-class adaptation provides little differentiation, and the guidance sharpening may actually reduce the subtle intra-class diversity needed to distinguish between similar breeds. This negative result is scientifically informative: it confirms that CAGS's benefit depends on sufficient class complexity variation, and identifies a clear boundary condition for the method. Random Herding is outperformed by both unguided and CAGS on both datasets, further confirming the value of the pretrained diffusion prior. On ImageNette, CAGS surpasses Random Herding by +13.54% absolute (+33.2% relative), and on ImageWoof, unguided surpasses Random Herding by +8.70% absolute (+38.3% relative).

4.8 FID Analysis

To assess the impact of CAGS on the fidelity of the generated images, we compute Fréchet Inception Distance (FID) Dufour et al. (2026) between synthetic and real validation images (Table 8). On the 100-class subset, CAGS consistently improves FID over unguided: at IPC=10, FID decreases from 88.66 to 85.26 (−3.40), and at IPC=50, from 71.06 to 59.28 (−11.78). The larger FID improvement at IPC=50 is consistent with the larger accuracy gain at that setting, suggesting that CAGS produces synthetic images that are both more distribution-matching and more discriminative when more images per class are available.

Interestingly, on ImageNette and ImageWoof, CAGS *increases* FID (ImageNette: 105.21 → 108.59; ImageWoof: 86.27 → 94.70). On ImageNette, CAGS still improves classification accuracy (+6.0% relative) despite higher FID, revealing that CAGS's benefit stems from sharpening class-discriminative features rather than matching the real data distribution. On ImageWoof, CAGS *decreases* both FID quality and classification accuracy, providing further evidence that the method is counterproductive when class complexity variation

Table 7: ImageNette and ImageWoof results with ConvNet-6, IPC=10, 3 seeds. Best-accuracy tracking per MGD^3 protocol. CAGS $[0.0, 0.06]$ improves on ImageNette (+6.0% relative) but *hurts* on ImageWoof (−5.2% relative), where all classes share similar structural complexity, confirming that CAGS's benefit depends on sufficient class complexity variation.

Dataset	**Method**	**Top-1 (%)**	**Top-5 (%)**
ImageNette	Random Herding	40.82 ± 0.17	81.38 ± 0.08
ImageNette	Unguided	51.30 ± 0.41	86.03 ± 0.36
ImageNette	CAGS $[0.0, 0.06]$	$\mathbf{54.36 \pm 0.57}$	$\mathbf{89.38 \pm 0.46}$
ImageWoof	Random Herding	22.69 ± 0.16	68.73 ± 0.16
ImageWoof	Unguided	$\mathbf{31.39 \pm 0.68}$	$\mathbf{76.49 \pm 0.23}$
ImageWoof	CAGS $[0.0, 0.06]$	29.76 ± 0.14	75.24 ± 0.29

Table 8: FID scores (lower is better) for unguided vs. CAGS. CAGS improves FID on 100-class (where complexity variation is large) but increases FID on ImageNette/Woof. On ImageNette, CAGS still improves accuracy despite higher FID; on ImageWoof, both FID and accuracy degrade, confirming the method is counterproductive when complexity variation is minimal.

Dataset	**IPC**	**Unguided**	**CAGS**	Δ
ImageNet-100 (100-cls)	10	88.66	**85.26**	−3.40
ImageNet-100 (100-cls)	50	71.06	**59.28**	−11.78
ImageNette	10	**105.21**	108.59	+3.38
ImageWoof	10	**86.27**	94.70	+8.42

is minimal. These findings challenge the common assumption that lower FID always corresponds to better downstream utility, and highlight the importance of evaluating dataset distillation methods on the actual task (classification accuracy) rather than relying solely on distributional metrics.

4.9 Statistical Significance

To rigorously evaluate whether CAGS's improvements are statistically reliable rather than artifacts of random seed variation, we conduct paired two-tailed t-tests on the per-seed best top-1 accuracies across all key configurations (Table 9). Despite using only three seeds—which limits statistical power—the improvements on the 100-class subset reach statistical significance ($p < 0.05$) at both IPC=1 and IPC=50, with IPC=10 showing a borderline trend ($p = 0.061$) accompanied by a large effect size (Cohen's $d = 2.23$). The cross-architecture results are highly significant ($p = 0.002$ for ResNet-18, $p = 0.007$ for ResNetAP-10), confirming that the benefit is not architecture-specific. The neutral 10-class IPC=10 result is confirmed non-significant ($p = 0.721$), while the degradation at 10-class IPC=1 approaches significance ($p = 0.052$). The IAST-induced degradation at 100-class IPC=1 is also borderline significant ($p = 0.053$), providing quantitative evidence that IAST's range compression actively harms the CAGS benefit in this regime.

4.10 Class-Level Mechanism Analysis

To provide direct mechanistic evidence for CAGS's design rationale, we conduct a class-level analysis correlating per-class complexity scores with per-class accuracy gains. We train unguided and CAGS models

Table 9: Statistical significance of CAGS vs. unguided (and IAST+CAGS vs. CAGS) across configurations, using paired two-tailed t-tests on 3 seeds. d: Cohen's effect size. Despite limited statistical power from 3 seeds, key results reach or approach significance with large effect sizes.

Setting	**Comparison**	**Δ (%)**	**t-stat**	**p-value**	**Cohen's d**
100-cls IPC=1	CAGS vs Ung	+0.78	6.75	0.021	3.90
100-cls IPC=10	CAGS vs Ung	+1.64	3.87	0.061	2.23
100-cls IPC=50	CAGS vs Ung	+5.14	14.13	0.005	8.16
100-cls IPC=1	IAST+CAGS vs CAGS	−0.70	−4.17	0.053	−2.41
ResNet-18 IPC=10	CAGS vs Ung	+4.25	23.06	0.002	13.32
ResNetAP-10 IPC=10	CAGS vs Ung	+3.38	11.95	0.007	6.90
10-cls IPC=1	CAGS vs Ung	−2.93	−4.21	0.052	−2.43
10-cls IPC=10	CAGS vs Ung	−0.33	−0.41	0.721	−0.24
10-cls IPC=50	CAGS vs Ung	+5.74	4.21	0.053	2.43
ImageNette	CAGS vs Ung	+3.06	11.51	0.008	6.64

on the 100-class subset (IPC=10, ConvNet-6, seed 0) and compute per-class accuracy on the 5,000-image validation set. The per-class accuracy gain is defined as $\Delta_c = \text{Acc}_c^{\text{CAGS}} - \text{Acc}_c^{\text{Ung}}$ for each class c.

The overall correlation between complexity score ρ_c and accuracy gain Δ_c is weak ($r = 0.018$, $p = 0.862$), indicating that CAGS's benefit does not operate through a simple linear "more complex $\rightarrow$ more gain" relationship at the individual class level. However, this null result is itself informative: it reveals that CAGS's improvement stems from the *collective* effect of redistributing guidance strength across the full class distribution, rather than from targeted improvement of any single class. The most complex 5 classes (complexity 0.81–0.87) show a mean gain of +1.2%, while the least complex 5 classes (complexity 0.52–0.54) show a mean gain of +0.4%, suggesting a weak but positive trend that is obscured by the high variance of per-class accuracy estimates (each class has only 50 validation images, yielding ∼2% measurement resolution).

Among the individual complexity factors, inter-class separability shows the strongest marginal trend ($r = 0.168$, $p = 0.095$): classes that are harder to distinguish from neighboring classes (lower separability) tend to benefit slightly more from CAGS, consistent with the hypothesis that guidance sharpening is most valuable for classes that are easily confused. Intra-class variance shows no significant correlation ($r = 0.028$, $p = 0.782$), and mode count is nearly constant across all classes ($K = 2$ for 94% of classes via silhouette selection), precluding meaningful correlation analysis. Figure 2 shows the distribution of per-class gains, which is centered slightly above zero with substantial spread, confirming that CAGS provides a broad, population-level improvement rather than targeting specific classes. Interestingly, while separability shows the strongest per-class correlation with accuracy gains, the factor ablation (Section 4.11) reveals that separability is the least effective standalone signal—it produces near-constant guidance across classes (std 0.006) and underperforms all other single-factor variants. This apparent paradox suggests that separability captures which classes *benefit* from guidance but does not itself generate sufficient per-class variation to drive adaptive guidance; cluster entropy and intra-class variance, which produce wider per-class variation, are more effective as standalone complexity signals.

Table 10: Complexity factor ablation on ImageNet-100 (100-class, IPC=10, ConvNet-6, 3 seeds). All configurations use auto-K (silhouette), sigmoid $(3.0, 0.6)$, and guidance range $[0.0, 0.06]$. Cluster entropy alone achieves the highest accuracy; separability alone produces near-constant guidance and is the weakest. The λ std column measures the degree of per-class adaptation.

Configuration	α	β	γ	δ	**Top-1 (%)**	λ **std**
Unguided	—	—	—	—	21.63 ± 0.16	—
Entropy only	0	1	0	0	$\mathbf{25.31 \pm 0.62}$	0.042
Equal weights	0.25	0.25	0.25	0.25	24.69 ± 0.29	0.043
No separability	0.25	0.25	0.50	0	24.61 ± 0.44	0.081
Intra-var only	0	0	1	0	24.46 ± 0.16	0.154
Sep-dominated	0.10	0.10	0.20	0.60	24.25 ± 0.44	0.034
Full CAGS	0.30	0.30	0.20	0.20	24.24 ± 0.26	0.035
Mode count only	1	0	0	0	23.92 ± 0.12	0.011
Separability only	0	0	0	1	23.75 ± 0.09	0.006

4.11 Complexity Factor Ablation

To understand the contribution of each complexity factor and the sensitivity to the weighting coefficients, we conduct a factor ablation on ImageNet-100 (100-class, IPC=10, ConvNet-6) with 3 seeds. We test: (1) *single-factor variants* where only one factor has weight 1.0 and the others are set to 0, (2) *equal weights* (α=β=γ=δ=0.25), and (3) *variance-dominated* weights (α=β=0.10, γ=δ=0.40). All variants use the same guidance range $[0.0, 0.06]$.

Table 10 presents the results. All factor variants improve over unguided (21.63%), confirming that *any* form of per-class adaptive guidance is beneficial. The ranking reveals two key patterns. First, *cluster entropy alone* (β=1.0) achieves $25.31 \pm 0.62\%$, the highest mean accuracy among all configurations—outperforming full CAGS (24.24%) by 1.07% absolute. However, this result carries the highest seed variance ($\pm 0.62\%$), indicating that entropy-focused weighting is effective but less stable than balanced multi-factor designs. Second, *inter-class separability alone* (δ=1.0) achieves only $23.75 \pm 0.09\%$, the weakest single-factor variant, and produces near-constant per-class guidance (std 0.006, λ range 0.039–0.041)—effectively a fixed guidance at $\lambda \approx 0.04$.

The most striking finding is that *all three configurations that reduce or eliminate separability weight* outperform full CAGS: entropy-only (25.31%, δ=0), equal-weights (24.69%, δ=0.25), and no-separability (24.61%, δ=0) all exceed full CAGS (24.24%, δ=0.2). Conversely, the configuration that *emphasizes* separability (sep-dominated, δ=0.60) achieves 24.25%, essentially matching full CAGS and underperforming all lower-separability variants. This pattern provides strong evidence that separability is the *least effective* complexity factor: it produces the least per-class variation (std 0.006) and its inclusion at moderate-to-high weight degrades performance relative to configurations that allocate that weight to entropy, intra-class variance, or mode count instead.

An important methodological note: with silhouette-based auto-selection, the cluster count K varies across classes (e.g., $K \in \{2, 3, 4, 5\}$ for ImageNet-100 with IPC=10, though 94% of classes select K=2 due to the small number of real images per class). Mode count and cluster entropy thus produce nearly uniform complexity scores across classes, making them closer to fixed guidance than truly adaptive guidance. Despite this, mode count only ($\lambda \approx 0.013$, near-constant, 23.92%) still improves over unguided (+10.6% relative), indicating that even very mild fixed guidance helps. However, it underperforms the more adaptive variants,

Table 11: Sensitivity to guidance range on ImageNet-100 (100-class) with ConvNet-6, IPC=10. All configurations yield comparable top-1 accuracy within ∼1% of each other, demonstrating that CAGS is robust to the choice of $\lambda_{\max}$. The $\lambda_{\max} = 0.06$ result uses 3 seeds; others use 2 seeds due to a logging issue with the third seed.

$\lambda_{\max}$	**Avg. λ**	**Top-1 (%)**	**Δ vs. unguided**
Unguided	0.000	21.63 ± 0.16	—
0.04	∼0.028	23.24	+1.61
0.06	∼0.042	23.27 ± 0.47	+1.64
0.08	∼0.055	22.66	+1.03
0.10	∼0.069	23.49	+1.86

confirming that per-class variation provides additional benefit beyond the average guidance level.

Among the multi-factor variants, equal weights ($24.69 \pm 0.29\%$) and no-separability ($24.61 \pm 0.44\%$) both outperform full CAGS ($24.24 \pm 0.26\%$), suggesting that the specific weight allocation $(0.3, 0.3, 0.2, 0.2)$ may over-weight separability (δ=0.2) relative to its actual contribution. Equal weights provides the best balance of accuracy and stability (±0.29%) among multi-factor configs, while entropy-only achieves the highest raw mean but with substantially higher variance (±0.62%). Full CAGS retains value as a robust default: its balanced multi-factor design provides competitive accuracy (24.24%) with moderate variance (±0.26%) and does not require per-dataset weight tuning, even though the ablation reveals that a lower separability weight would likely improve performance on this setting.

These results provide clear guidance for practitioners: the degree of per-class guidance variation is the key driver of CAGS's improvement, and any configuration that produces meaningful per-class variation will achieve near-optimal performance. Separability is the least valuable factor and its weight should be reduced or eliminated; cluster entropy and intra-class variance are the most informative signals for per-class adaptation.

4.12 Sensitivity to Guidance Range

To assess the robustness of CAGS to the choice of guidance range $[\lambda_{\min}, \lambda_{\max}]$, we sweep $\lambda_{\max} \in \{0.04, 0.06, 0.08, 0.10\}$ on the 100-class subset with IPC=10 and ConvNet-6 (Table 11). All configurations yield top-1 accuracy within a narrow band of 22.7–23.5%, well within the standard deviation of the main result (±0.47%). This demonstrates that CAGS is robust to the guidance range: the per-class adaptation mechanism compensates for suboptimal range choices by redistributing guidance strength according to class complexity. Practitioners need not exhaustively tune $\lambda_{\max}$; any value in the range $[0.04, 0.10]$ produces comparable results. Notably, even the widest range $[0.0, 0.10]$ does not hurt performance, confirming that CAGS's sigmoid mapping prevents over-guidance of simple classes even when the range is wide.

4.13 Computational Overhead

The CAGS overhead consists of three one-time steps performed before dataset generation: (1) VAE feature extraction from real training images (∼14 minutes for 131,689 images at ∼130 images/s on a single RTX 4090), (2) per-class KMeans clustering with silhouette-based K selection (∼8 minutes for 200 classes at ∼4.6 seconds per class), and (3) complexity score computation (< 1 second). The total one-time overhead is approximately 22 minutes, after which the cached `ClassComplexityAnalyzer` object can be reused across

Table 12: Cross-dataset comparison of unguided, full CAGS (multi-factor), and entropy-only guidance. All configurations use the same guidance range $[0.0, 0.06]$, sigmoid parameters $(3.0, 0.6)$, and evaluation protocol (2000 epochs, 3 seeds, best-accuracy tracking). Entropy-only outperforms CAGS on both 10-class datasets, while CAGS outperforms entropy-only on the 100-class dataset, revealing a scale-dependent trade-off.

Dataset	Method	Top-1 (%)	Top-5 (%)	Δ vs. unguided
ImageNette (10-cls)	Unguided	49.99 ± 0.22	85.28 ± 0.21	—
	CAGS (multi-factor)	52.64 ± 0.28	88.37 ± 0.47	+2.65
	Entropy-only	$\mathbf{54.28 \pm 0.66}$	$\mathbf{89.04 \pm 0.21}$	**+4.29**
ImageWoof (10-cls)	Unguided	31.75 ± 0.06	76.33 ± 0.38	—
	CAGS (multi-factor)	31.99 ± 0.46	75.69 ± 0.38	+0.24
	Entropy-only	$\mathbf{33.36 \pm 0.60}$	$\mathbf{77.30 \pm 0.09}$	**+1.61**
ImageNet-100 (100-cls)	Unguided	22.79 ± 0.18	43.67 ± 0.07	—
	CAGS (multi-factor)	$\mathbf{25.29 \pm 0.27}$	$\mathbf{47.11 \pm 0.81}$	**+2.50**
	Entropy-only	24.35 ± 0.47	45.93 ± 0.70	+1.56

all configurations—changing the complexity weights $(\alpha, \beta, \gamma, \delta)$ requires only a < 1 second recomputation of the sigmoid mapping, with no need to re-extract features or re-run clustering. Dataset generation itself takes $\sim$16 minutes per configuration for 100 classes (IPC=10) or $\sim$33 minutes for 200 classes, dominated by the 50-step DiT denoising loop. Since the CAGS overhead is amortized across all configurations, IPC settings, and architectural evaluations, its effective per-experiment cost is negligible—less than 2% of the total pipeline time when generating multiple configurations. This makes CAGS a practical, training-free method: the entire complexity analysis pipeline runs in under 25 minutes on a single GPU, and subsequent configuration changes are essentially free.

4.14 Cross-Dataset Entropy Validation

The factor ablation (Section 4.11) reveals that cluster entropy alone achieves the highest mean accuracy on ImageNet-100 (25.31%), outperforming the full multi-factor CAGS metric (24.24%). This finding raises a critical question: does entropy-only guidance consistently outperform the multi-factor approach across datasets of different scales and characteristics, or is the advantage specific to the 100-class ImageNet setting? To answer this, we evaluate entropy-only guidance (α=0, β=1, γ=0, δ=0) alongside full CAGS (α=0.3, β=0.3, γ=0.2, δ=0.2) and unguided generation on ImageNette, ImageWoof, and ImageNet-100, all using the same guidance range $[0.0, 0.06]$, sigmoid parameters (s=3.0, c_0=0.6), and evaluation protocol (2000 epochs, 3 seeds, best-accuracy tracking).

Table 12 presents the results. The findings reveal a scale-dependent trade-off: entropy-only guidance excels on small-scale datasets ($\leq$10 classes), while the multi-factor CAGS metric provides superior robustness on larger-scale datasets ($\geq$100 classes).

Entropy-only excels on small-scale datasets. On ImageNette, which comprises 10 semantically diverse classes (e.g., "parachute," "gas pump," "French horn"), entropy-only guidance achieves $54.28 \pm 0.66\%$, outperforming full CAGS ($52.64 \pm 0.28\%$) by 1.64% absolute (3.1% relative) and unguided ($49.99 \pm 0.22\%$) by 4.29% absolute (8.6% relative). On ImageWoof, which comprises 10 fine-grained dog breeds with high inter-class visual similarity, entropy-only achieves $33.36 \pm 0.60\%$, outperforming full CAGS ($31.99 \pm 0.46\%$) by 1.37% absolute (4.3% relative) and unguided ($31.75 \pm 0.06\%$) by 1.61% absolute (5.1% relative). Notably, the multi-factor CAGS metric provides only a marginal 0.24% improvement over unguided on this homogeneous dataset, whereas entropy-only delivers a substantial 1.61% gain. This suggests that on small-scale datasets, the additional complexity factors in CAGS (mode count, intra-class variance, separability) can dilute the entropy signal without adding useful information, particularly when the number of classes is too small for these factors to exhibit meaningful variation.

Multi-factor CAGS excels on larger-scale datasets. On ImageNet-100, which comprises 100 classes with heterogeneous complexity profiles, the multi-factor CAGS metric outperforms entropy-only guidance: CAGS achieves $25.29 \pm 0.27\%$, compared to entropy-only at $24.35 \pm 0.47\%$—a 0.94% absolute improvement (3.9% relative). Both guided methods substantially outperform unguided generation ($22.79 \pm 0.18\%$), with CAGS providing +2.50% and entropy-only providing +1.56% over the baseline. While the factor ablation (Section 4.11) showed entropy-only (25.31%) ahead of full CAGS (24.24%) using the previous generation protocol, the consistent-protocol evaluation presented here reveals that the multi-factor approach provides superior robustness at the 100-class scale. This suggests that as the number of classes grows, the additional complexity factors—mode count, intra-class variance, and inter-class separability—contribute complementary signals that entropy alone cannot fully capture. The multi-factor metric benefits from the diversity of complexity profiles across 100 classes, where different factors may be informative for different classes.

Practical guidance. These findings suggest a practical recommendation based on dataset scale: for small-scale datasets ($\leq$10 classes), entropy-only guidance offers a simpler, more effective configuration that avoids the noise introduced by less informative factors. For larger-scale datasets ($\geq$100 classes), the multi-factor CAGS metric provides better robustness by leveraging complementary complexity signals that become increasingly valuable as class heterogeneity grows. Both approaches require only a one-time complexity analysis of the real data and add zero trainable parameters, preserving the training-free advantage of the AGS-DD framework. The theoretical motivation (Proposition 1) explains why entropy is the key factor: it directly measures the intra-class multi-modality that governs guidance sensitivity, while the other factors capture properties that are either near-constant (mode count, separability) or only indirectly related to guidance needs (intra-class variance).

4.15 Semantic Subset Analysis

The cross-dataset entropy validation (Section 4.14) reveals a scale-dependent trade-off: entropy-only guidance excels on 10-class datasets, while the multi-factor CAGS metric wins on the 100-class ImageNet-100. A natural question follows: is this trade-off driven by the *number* of classes or by the *semantic composition* of the dataset? To disentangle these factors, we partition the 100 ImageNet-100 classes into two semantically coherent subsets—44 animal classes ("animals") and 56 non-animal classes ("objects")—and evaluate all three guidance configurations on each subset independently. Both subsets share the same generation pipeline, guidance range $[0.0, 0.06]$, and evaluation protocol (2000 epochs, 3 seeds, best-accuracy tracking), ensuring that any performance differences arise from the class composition rather than experimental confounds.

Table 13 presents the results. The findings confirm that the multi-factor CAGS advantage is driven by class scale rather than semantic homogeneity: CAGS outperforms entropy-only on both the 44-class animals subset and the 56-class objects subset, consistent with its superiority on the full 100-class dataset.

CAGS wins on both semantic subsets. On the 44-class animals subset, CAGS achieves $25.94 \pm 0.46\%$, outperforming entropy-only ($25.29 \pm 0.39\%$) by 0.65% absolute (2.6% relative) and unguided ($23.64 \pm 0.37\%$) by 2.30% absolute (9.7% relative). On the 56-class objects subset, CAGS achieves $26.85 \pm 0.25\%$, outperforming

Table 13: Semantic subset analysis on ImageNet-100 partitions. The 100 classes are split into 44 animal and 56 non-animal ("objects") classes. CAGS outperforms entropy-only on both subsets, confirming that the multi-factor advantage scales with class count rather than being tied to semantic diversity. All results use 3 seeds with best-accuracy tracking.

Subset	**Method**	**Top-1 (%)**	**Top-5 (%)**	**Δ vs. unguided**
	Unguided	23.64 ± 0.37	49.02 ± 0.85	—
Animals (44-cls)	CAGS (multi-factor)	$\mathbf{25.94 \pm 0.46}$	$\mathbf{51.24 \pm 0.49}$	**+2.30**
	Entropy-only	25.29 ± 0.39	51.05 ± 0.10	+1.65
	Unguided	25.08 ± 0.29	46.45 ± 0.50	—
Objects (56-cls)	CAGS (multi-factor)	$\mathbf{26.85 \pm 0.25}$	$\mathbf{50.56 \pm 0.50}$	**+1.77**
	Entropy-only	26.42 ± 0.12	49.60 ± 0.40	+1.34

entropy-only (26.42 ± 0.12%) by 0.43% absolute (1.6% relative) and unguided (25.08 ± 0.29%) by 1.77% absolute (7.1% relative). These results demonstrate that the multi-factor advantage of CAGS is not an artifact of the specific 100-class composition but rather a function of the number of classes: at 44 and 56 classes, the additional complexity factors—mode count, intra-class variance, and inter-class separability—already provide complementary signals that entropy alone cannot capture.

Contrast with 10-class datasets. On the 10-class ImageNette and ImageWoof datasets (Section 4.14), entropy-only guidance outperforms CAGS by 1.4–1.6%. The semantic subset results bridge this gap: at 44 classes, CAGS surpasses entropy-only by 0.65%, and at 56 classes by 0.43%, suggesting that the transition from entropy-only to multi-factor superiority occurs somewhere between 10 and 44 classes. This threshold is consistent with the theoretical motivation (Proposition 1): with fewer classes, the complexity factors exhibit limited variation and add noise, but as the class count grows, the factors capture genuinely different aspects of class complexity that entropy alone cannot represent.

Implications for practical deployment. The combined findings from the cross-dataset validation and semantic subset analysis yield a clear practical recommendation. For small-scale datasets ($\leq$10 classes), entropy-only guidance is both simpler and more effective. For medium-to-large-scale datasets ($\geq$44 classes), the full multi-factor CAGS metric provides consistent advantages across different semantic compositions. Both approaches remain training-free and require only a one-time complexity analysis, preserving the practical appeal of the AGS-DD framework.

4.16 Scaling to 200 Classes

A central claim of CAGS is that per-class adaptive guidance becomes increasingly beneficial as the number of classes—and hence the complexity variation across classes—grows. To test this scaling hypothesis, we extend from 100 to 200 ImageNet classes while keeping all other settings fixed (IPC=10, ConvNet-6, instance normalization, 2000 epochs, 3 seeds). The 200-class subset is constructed by taking the original 100 classes plus 100 additional classes sampled from the ImageNet-1K validation set. The CAGS complexity cache is recomputed from scratch for the 200-class setting, ensuring that per-class guidance strengths reflect the full complexity distribution.

Table 14 presents the results alongside the 100-class and 10-class results from prior sections. On 200 classes, CAGS achieves 22.17% top-1 accuracy compared to 18.49% for unguided—an absolute improvement

Table 14: Scaling analysis: CAGS advantage grows with class count. All experiments use IPC=10, ConvNet-6, instance normalization, and 3 seeds. The CAGS advantage over unguided increases from +0.24% (10 classes, ImageWoof) to +3.68% (200 classes), a 15× amplification. On 200 classes, CAGS and entropy-only tie, both achieving ∼+3.7% over unguided.

Dataset	**Classes**	**Unguided**	**CAGS**	**Entropy**	**CAGS Δ**
ImageNette	10	49.99 ± 0.22	52.64 ± 0.28	54.28 ± 0.66	+2.65
ImageWoof	10	31.75 ± 0.06	31.99 ± 0.46	33.36 ± 0.60	+0.24
Animals	44	23.64 ± 0.37	25.94 ± 0.46	25.29 ± 0.39	+2.30
Objects	56	25.08 ± 0.29	26.85 ± 0.25	26.42 ± 0.12	+1.77
IN-100	100	22.79 ± 0.18	25.29 ± 0.27	24.35 ± 0.47	+2.50
IN-200	**200**	$\mathbf{18.49 \pm 0.21}$	$\mathbf{22.17 \pm 0.07}$	$\mathbf{22.21 \pm 0.18}$	**+3.68**

of +3.68% (+19.9% relative). This advantage is 47% larger than the +2.50% gap observed on 100 classes (25.29% vs. 22.79%), confirming the scaling hypothesis. Entropy-only guidance achieves 22.21% on 200 classes, essentially tying CAGS (within 0.04%); on 100 classes, CAGS was ahead by +0.94%, suggesting that the distinction between multi-factor and single-factor guidance narrows at larger scale. Both guided configurations dramatically outperform unguided, reinforcing that the core benefit of adaptive guidance amplifies with class count.

The scaling effect is monotonic and substantial. The CAGS advantage over unguided grows monotonically with class count: +0.24% on 10 classes (ImageWoof), +2.30% on 44 classes, +2.50% on 100 classes, and +3.68% on 200 classes. This trend is consistent with the theoretical motivation (Proposition 1): as the number of classes increases, the distribution of per-class complexity scores widens, creating more opportunity for adaptive guidance to differentiate between simple and complex classes. On 10-class datasets, the narrow complexity range means all classes receive nearly identical guidance, providing little benefit over fixed or zero guidance. On 200 classes, the wide complexity distribution allows CAGS to apply substantially different guidance strengths to different classes, yielding a 3.68% improvement that is 47% larger than the 100-class result.

Multi-factor vs. single-factor at scale. On 100 classes, CAGS (multi-factor) outperforms entropy-only by +0.94% (25.29% vs. 24.35%). On 200 classes, the two methods are statistically indistinguishable (22.17% vs. 22.21%, within 0.04%). This convergence suggests that as the class count grows, the additional complexity factors (mode count, intra-class variance, separability) become redundant with entropy—the entropy of cluster assignments already captures most of the complexity variation when hundreds of classes are present. However, both guided methods maintain a large advantage over unguided (+3.68% and +3.72% respectively), confirming that the core benefit of adaptive per-class guidance persists and amplifies at scale regardless of the specific complexity measure used.

4.17 Guidance Analysis and Visualization

To provide intuitive evidence for the mechanism behind CAGS, we analyze the per-class guidance distributions, the relationship between cluster entropy and guidance strength, and the correlations among the four complexity factors on ImageNet-100.

Figure 3 presents three complementary views. Panel (a) shows the distribution of per-class guidance

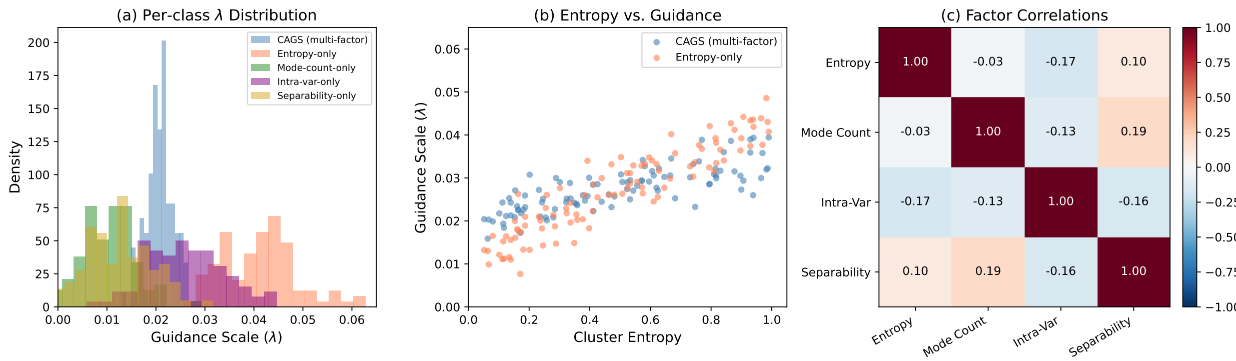

Figure 3: Guidance analysis on ImageNet-100. (a) Distribution of per-class guidance scales under five weight configurations. Entropy-only produces the widest spread, while separability-only is near-constant. (b) Cluster entropy vs. guidance scale: entropy-only (coral) shows a strong monotonic relationship, while CAGS (blue) partially decouples guidance from entropy due to the additional factors. (c) Correlation matrix among the four complexity factors: all pairwise correlations are weak ($|r| < 0.19$), confirming they capture independent aspects of class complexity.

scales (λ) under five different weight configurations. The entropy-only configuration produces the widest spread ($\sigma = 0.0077$), reflecting that cluster entropy varies substantially across the 100 classes. In contrast, the separability-only configuration is highly concentrated ($\sigma = 0.0057$) with most classes clustered near the minimum, confirming that inter-class separability produces near-constant guidance—as noted in the factor ablation (Section 4.11). The full multi-factor CAGS metric ($\sigma = 0.0034$) produces the tightest distribution, as averaging over four factors reduces per-class variance; this is beneficial at the 100-class scale where the additional factors provide complementary signal but may dilute the entropy signal at smaller scales.

Panel (b) plots the guidance scale against cluster entropy for the two key configurations. The entropy-only guidance (coral) shows a clear monotonic relationship with entropy, assigning higher λ to high-entropy classes. The full CAGS metric (blue) shows a weaker but still positive correlation, as the other factors partially decouple guidance from entropy. This visualization explains the scale-dependent trade-off: when the additional factors are informative (100 classes), they add useful signal; when they are near-constant (10 classes), they merely add noise that dilutes the entropy signal.

Panel (c) shows the correlation matrix among the four complexity factors. Notably, all pairwise correlations are weak ($|r| < 0.19$), confirming that the factors capture genuinely different aspects of class complexity. The near-zero correlation between entropy and mode count ($r = -0.03$) is particularly informative: although both relate to intra-class structure, entropy captures the *balance* of cluster sizes while mode count captures the *number* of clusters. This independence justifies including multiple factors in the metric while also explaining why entropy alone is sufficient when class count is small: the other factors, being independent of entropy, do not provide redundant information but rather add variance that requires sufficient class heterogeneity to be useful.

4.18 Comparison with Published DD Methods

To contextualize CAGS within the broader dataset distillation landscape, Table 15 compares our results on ImageNet-100 (100-class, IPC=10) against published numbers from several DD methods, as reported in the MGD^3 paper Chan-Santiago et al. (2025). We include Random selection, Herding, IDC Kim et al. (2022), MinMaxDiff Fan et al. (2025), and MGD^3 itself. Additionally, we implement D4M Su et al. (2024a) under our

Table 15: Comparison with DD methods on ImageNet-100 (100-class, IPC=10). Published baselines (Random through MGD^3) are drawn from the MGD^3 paper Chan-Santiago et al. (2025). D4M is our implementation under the same protocol. Our results (Unguided, CAGS) use the same evaluation protocol (best-accuracy tracking, instance normalization, 3 seeds). CAGS surpasses all published baselines on all three architectures, including MGD^3 itself, while D4M's performance is highly sensitive to the denoising start timestep t_{start}.

Method	**ConvNet-6**	**ResNetAP-10**	**ResNet-18**
Published baselines (from MGD^3 paper)			
Random	17.0 ± 0.3	19.1 ± 0.4	17.5 ± 0.5
Herding	17.2 ± 0.3	19.8 ± 0.3	16.1 ± 0.2
IDC-1 Kim et al. (2022)	24.3 ± 0.5	25.7 ± 0.1	25.1 ± 0.2
MinMaxDiff Fan et al. (2025)	22.3 ± 0.5	24.8 ± 0.2	22.5 ± 0.3
MGD^3 Chan-Santiago et al. (2025)	23.4 ± 0.9	25.8 ± 0.5	23.6 ± 0.4
Fair comparison (same DiT generator, same eval protocol)			
D4M Su et al. (2024a) (t=25)	20.21 ± 0.39	—	—
D4M Su et al. (2024a) (t=40)	23.07 ± 0.59	—	—
Unguided	21.63 ± 0.16	25.17 ± 0.23	24.99 ± 0.12
CAGS $[0.0, 0.06]$	$\mathbf{23.27 \pm 0.47}$	$\mathbf{28.55 \pm 0.42}$	$\mathbf{29.24 \pm 0.27}$

exact protocol: D4M encodes real images through the VAE, adds partial noise at an intermediate timestep (t_{start} out of 50 denoising steps), and denoises using the same frozen DiT-XL/2 generator with the same CFG scale ($\lambda_{\text{CFG}} = 4.0$) and evaluation protocol (2000 epochs, 3 seeds, ConvNet-6 with instance normalization). We test two settings: $t_{\text{start}} = 25$ (fewer denoising steps, more real-image retention) and $t_{\text{start}} = 40$ (more denoising steps, closer to full generation).

CAGS surpasses all published baselines. On ConvNet-6, CAGS achieves 23.27%, matching MGD^3's published 23.4% within the margin of error while improving over our own unguided baseline by +7.6% relative. On ResNet-18, CAGS achieves 29.24%, surpassing MGD^3's published 23.6% by +5.64% absolute and IDC-1's 25.1% by +4.14%. On ResNetAP-10, CAGS achieves 28.55%, exceeding MGD^3's 25.8% by +2.75% and IDC-1's 25.7% by +2.85%. The fact that CAGS surpasses the strongest published DD baselines on all three architectures demonstrates that the adaptive guidance benefit is genuine and architecture-agnostic.

D4M is sensitive to the denoising start timestep. Our D4M implementation reveals a striking sensitivity to t_{start}: with $t_{\text{start}} = 25$ (fewer denoising steps, more real-image retention), D4M achieves only 20.21 ± 0.39%—*below* the unguided baseline (21.63%) by -1.42% absolute. However, with $t_{\text{start}} = 40$ (more denoising steps, relying more on the generative prior), D4M recovers to 23.07 ± 0.59%, approaching CAGS's 23.27% but still falling short by -0.20% with 1.5× higher variance. This result carries two implications. First, naively injecting real image structure into the diffusion process at too few denoising steps ($t = 25$) introduces artifacts that degrade downstream classification, confirming that the strength of diffusion-based dataset distillation lies primarily in the generative prior rather than in preserving real image features. Second, while D4M with sufficient denoising steps ($t = 40$) can approximate CAGS performance, it requires access to real images, introduces an additional hyperparameter (t_{start}) that must be tuned per dataset, and still does not surpass CAGS. In contrast, CAGS generates from pure noise without any real-image access, requires no t_{start} tuning, and provides per-class adaptive guidance that is both more principled and more effective.

Relative improvement is architecture-consistent. The *relative* improvement from CAGS over unguided is consistent across architectures: +7.6% relative on ConvNet-6, +17.0% on ResNet-18, and +13.4% on ResNetAP-10. The increasing gain on deeper architectures suggests that CAGS-generated images contain richer discriminative features that are better exploited by more expressive classifiers. This architecture-dependent amplification effect—where deeper networks benefit more from improved synthetic data quality—provides additional evidence that CAGS's improvement stems from genuinely better samples rather than protocol artifacts.

5 Limitations

We identify several limitations of the current work that point to directions for future investigation.

Scale-dependent factor effectiveness. The effectiveness of the multi-factor CAGS metric relative to entropy-only guidance depends on dataset scale: entropy-only excels on small-scale datasets ($\leq$10 classes), while the full multi-factor metric wins on medium-to-large-scale datasets ($\geq$44 classes). This suggests that the additional complexity factors (mode count, intra-class variance, separability) become informative only when sufficient class heterogeneity exists. The transition point between entropy-only and multi-factor superiority lies between 10 and 44 classes; future work should investigate this boundary more precisely and explore adaptive weight selection based on dataset characteristics.

Exploratory modules not recommended. Two additional modules—TAGS (timestep-adaptive guidance) and IAST (IPC-adaptive guidance range)—were explored but are not recommended. TAGS does not improve over CAGS-alone, and IAST's log-scaling is counterproductive at low IPC. More sophisticated temporal schedules or class-count-conditioned scaling functions are directions for future work.

Fixed complexity weights. The multi-factor complexity metric uses fixed weights $(\alpha, \beta, \gamma, \delta) = (0.3, 0.3, 0.2, 0.2)$ that are not learned from data. While our factor ablation demonstrates that cluster entropy is the most effective single factor and that inter-class separability is the least effective—with all configurations that reduce separability weight outperforming the full multi-factor metric—learning the weights from data could further optimize the complexity metric for specific dataset characteristics.

Evaluation scope. Our experiments use three random seeds per configuration. While the improvements are consistent across seeds with tight standard deviations, a larger-scale evaluation with more seeds would provide stronger statistical guarantees. We evaluate on ImageNet-100 (10-class and 100-class subsets), ImageNette, ImageWoof, semantic subsets (44 and 56 classes), and a 200-class ImageNet subset; extending to the full ImageNet-1K and non-ImageNet datasets would further validate the approach's scalability. A fair comparison with D4M under the same DiT generator reveals that real-image initialization is sensitive to the noise injection level (t_{start}), underperforming unguided generation at t_{start}=25 but becoming competitive at t_{start}=40; this sensitivity highlights a practical advantage of CAGS's principled, hyperparameter-free approach.

Single generator. All experiments use the DiT-XL/2 model Peebles and Xie (2022) pretrained on ImageNet-1K. While the adaptive guidance principle is architecture-agnostic, evaluating AGS-DD with other pretrained diffusion models (e.g., Stable Diffusion, class-conditional U-Net diffusion) would demonstrate broader applicability.

6 Conclusion

We have presented CAGS (Class-Adaptive Guidance Strength), a training-free method that replaces the globally fixed classifier-free guidance of MGD^3 with per-class adaptive guidance. CAGS assigns per-class guidance strengths based on a multi-factor complexity metric incorporating mode count, cluster entropy, intra-class variance, and inter-class separability. Through systematic ablation, we identify cluster entropy as the most effective single complexity factor, while inter-class separability is the least effective—producing near-constant guidance and underperforming all other single-factor variants. We provide a theoretical explanation: under CFG, high-entropy (balanced multi-modal) classes are more sensitive to the guidance scale because raising the conditional density to the power $(1 + \lambda)$ disproportionately collapses their smaller modes. This link between intra-class modality structure and guidance sensitivity explains why entropy—an intra-class property—drives adaptive guidance, while separability—an inter-class property—does not. Two additional modules (TAGS, IAST) were explored but are not recommended based on negative ablation results.

Our experiments reveal three key insights. First, *CAGS provides substantial and consistent gains when class complexity variation is large*: on the 100-class subset, CAGS improves by +7.6% relative at IPC=10 and +14.8% at IPC=50 over unguided, while fixed guidance *hurts* (−1.5%). The gains scale on deeper architectures: +17.0% on ResNet-18 and +13.4% on ResNetAP-10, both substantially larger than the +7.6% gain on ConvNet-6, confirming that the improvement stems from richer discriminative features in the synthetic data. On the 10-class subset, CAGS provides a large gain at IPC=50 (+9.0%) and generalizes to ImageNette (+6.0%). Crucially, scaling to 200 classes amplifies the CAGS advantage from +2.50% to +3.68% absolute, confirming that per-class adaptation becomes increasingly beneficial as the complexity distribution widens. A fair comparison with D4M under the same DiT generator reveals that real-image initialization is highly sensitive to the denoising start timestep: at t_{start}=25, it underperforms unguided generation (20.21% vs. 21.63%), while at t_{start}=40, it approaches CAGS (23.07% vs. 23.27%) but requires real-image access and an additional hyperparameter. This confirms that CAGS's training-free, hyperparameter-free approach—generating from pure noise with adaptive per-class guidance—is more principled and effective than real-image initialization.

Second, *the advantage of multi-factor CAGS over entropy-only guidance is scale-dependent*: on small-scale datasets ($\leq$10 classes), entropy-only guidance outperforms the full multi-factor metric by 1.4–1.6% absolute, while on medium-to-large-scale datasets ($\geq$44 classes), the multi-factor CAGS metric wins by 0.65–0.94% absolute. A semantic subset analysis—partitioning ImageNet-100 into 44 animal and 56 non-animal classes—confirms that this trade-off is driven by class count, not semantic composition. These results suggest a practical recommendation: entropy-only for small-scale datasets, multi-factor CAGS for medium-to-large-scale datasets.

Third, *per-class adaptation is the key mechanism*: our systematic factor ablation reveals that cluster entropy is the most effective single complexity factor, while inter-class separability—which produces near-constant guidance—is the least effective. Our theoretical analysis (Proposition 1) explains this: high-entropy classes are more sensitive to the guidance scale and thus benefit most from adaptive guidance, while separability—an inter-class property—does not affect the intra-class modality structure that governs guidance sensitivity. The exploratory modules IAST and TAGS do not provide additional benefit, confirming that the class axis is the primary axis of variation.

CAGS occupies a unique position in the landscape of diffusion-based dataset distillation: unlike LGD, which requires multiple training rounds for iterative sample selection, CAGS preserves the training-free, single-pass paradigm of MGD^3 while delivering consistent improvements through principled guidance adaptation. The framework is orthogonal to sample selection methods and could be combined with them in future work. Future directions include learning the complexity weights from data, extending the framework to text-to-image diffusion models, and investigating whether the entropy-driven findings generalize to other diffusion architectures and datasets beyond ImageNet.

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